LESSON PLAN

AVIONICS AND ELECTRONIC WARFARE (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	25
Total No. of Lesson Plans	25	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Practical on operation on DC motor, Induction Motor (1Ø & 3 Ø), universal motor on AEMT
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Practical on operation on DC motor, Induction Motor (1Ø & 3 Ø), universal motor on AEMT.
5. Specific Objectives	13.1.4.1 knowledge on Safety precautions & Practical on cleaning of connecters, terminals.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Operation on DC, 1Ø Induction, 3Ø Induction, and Universal motors on AEMT; Cleaning of connectors and terminals.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the three types of motors covered in this practical?
(b) List two essential safety precautions before starting any motor operation.
(c) Why is cleaning connectors and terminals critical for motor performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Safety Precautions for AEMT Operation • Cleaning Connectors and Terminals • DC Motor Operation Procedure • Universal Motor Operation Procedure	L & Ds	PPT, IB	15 min
(b)	• Single Phase Induction Motor Operation • Three Phase Induction Motor Operation • Troubleshooting Common Motor Faults • Verification of Motor Rotation Direction	L & Ds	PPT, PDF	20 min
(c)	• Final Inspection of Motors • Documentation of Practical Results • Review of Safety Compliance • Q&A Session on Motor Operations	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Describe the correct sequence for cleaning motor terminals before operation.
(b) Explain the difference in starting procedures between a DC motor and a 3Ø Induction motor.
(c) Identify three signs that indicate a motor is overheating during operation.

12. Home Assignment and Follow Up	Prepare a brief report detailing the specific safety checks performed on the AEMT motors and how they relate to the safety protocols discussed in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	End of subject	Refer to the master syllabus reference list	-

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Instructor, Senior Instructor, P No. 12345	Senior Instructor, Chief Instructor, P No. 67890
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min